Karolina Finc

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Research Interests

In my research, I seek to understand how the organization and dynamics of complex networks shape human experience. I study how brain networks reorganize during cognition, learning, and flow states, and how the brain dynamically interacts with the body. I am particularly interested in connecting ideas across disciplines and exploring unexpected links between seemingly distant fields.

Experience

2020 – present	**Assistant Professor**, Institute of Advanced Studies, Centre for Modern Interdisciplinary Technologies, Nicolaus Copernicus University in Toruń, Poland.
2026 – present	**Member**, Human Ecology Lab, Institute of Advanced Studies, CMIT, Nicolaus Copernicus University in Toruń, Poland.
2020 – present	**Member**, Centre of Excellence Dynamics, Mathematical Analysis & Artificial Intelligence group: Neuroinformatics and Artificial Intelligence.
2025	**Visiting Researcher**, RITMO Centre for Interdisciplinary Studies in Rhythm, Time and Motion, University of Oslo, Norway. Collaboration on network analysis of musician–audience synchronization and autonomic regulation during live performances.
2021	**Postdoctoral Fellow**, Max Planck Institute for Human Development, Max Planck Research Group NeuroCode (advisor: Nicolas Schuck), Berlin, Germany. Contributed a network neuroscience perspective to research on memory and learning.
2019	**Visiting Researcher**, Stanford University, Department of Psychology, The Poldrack Lab (advisor: Russell Poldrack), Stanford, USA. Contributing to fMRIPrep and FitLins; representation similarity analysis of the self-regulation ontologies project.
2018	**Visiting Researcher**, University of Pennsylvania, Department of Bioengineering, Complex Systems Group (advisor: Danielle Bassett), Philadelphia, USA. Dynamic multilevel network analysis of longitudinal fMRI data.
2017	**Data Scientist**, Neurodio spin-off. Analysis of behavioral data from a cognitive training study.
2014 – present	**Co-founder**, Neurodio spin-off company focusing on meaningful design.
2014 – 2020	**Research Assistant**, Neurocognitive Laboratory, CMIT, Nicolaus Copernicus University in Toruń, Poland.
2014	**Visiting Student Researcher**, Max Planck Institute for Human Development, Center for Lifespan Psychology, Mechanisms and Sequential Progression of Plasticity Group (advisor: Simone Kühn), Berlin, Germany.
2014	**Intern**, Neurocognitive Laboratory, CMIT, Nicolaus Copernicus University in Toruń, Poland.

Education

2017 – 2019	**PhD, Natural Sciences in Physical Sciences** (16.10.2019), Faculty of Physics, Astronomy and Informatics, Nicolaus Copernicus University, Toruń. Thesis: *Dynamics and plasticity of human brain functional network during working memory task performance* (advisor: Włodzisław Duch).
2017 – 2018	**Applied Mathematics**, Faculty of Mathematics and Computer Science, Nicolaus Copernicus University.
2012 – 2014	**MA, Cognitive Science** (03.07.2014), Faculty of Humanities, Institute of Philosophy, Nicolaus Copernicus University. Thesis: *Effect of action video game training on temporal information processing in the range of tens of milliseconds.*

2009 – 2012 **BA, Cognitive Science** (04.07.2012), Faculty of Humanities, Institute of Philosophy, Nicolaus Copernicus University. Thesis: *The role of sleep in neuroplasticity in the learning and memory context.*

Publications

[1] Finc, K., Adamska-Stolarczyk, I., & Bassett, D. S. (2026). Flexible Modularity in the Human Brain: How Network Architecture Reconfigures Over Time. *Neuroscience & Biobehavioral Reviews*, 188, 106784.

[2] Bonna, K., Hulme, O. J., Steinkamp, S. R., Meder, D., van der Weij, M. E. C., Duch, W., & Finc, K. (2026). Brain network reconfiguration during reward prediction error processing. *Network Neuroscience* (accepted).

[3] Lewartowski, A. B., & Finc, K. (2026). Exploring the Potential of Polyrhythmic Entrainment for Cognitive Enhancement and Adaptive Brain-Body Dynamics. *Musicae Scientiae* (in press).

[4] Grassia, M., d'Andrea, V., Finc, K., De Domenico, M., & Mangioni, G. (2026). Machine-enhanced reconstruction of functional connectomes unravels discriminative brain sub-systems in health and disease. *Scientific Reports* (in press).

[5] Poetto, S., Merritt, H., Santoro, A., Rabuffo, G., Finc, K., Battaglia, D., ... & Petri, G. (2025). The topological architecture of brain identity. *bioRxiv*, 2025-06.

[6] Gut, M., Bonna, K., Finc, K., Wypych, A., Mańkowska, K., & Słupczewski, J. (2025). Transfer of the effects of computer-based cognitive training in non-numerical–spatial association to numerical–spatial association in children at risk of dyscalculia. *International Journal of Psychophysiology*, 213, 113037.

[7] Adamska, I., & Finc, K. (2023). Effect of LSD and music on the time-varying brain dynamics. *Psychopharmacology*, 240(7), 1601–1614.

[8] Niso, G., Botvinik-Nezer, R., Appelhoff, S., De La Vega, A., Esteban, O., Etzel, J. A., Finc, K., ... & Rieger, J. (2022). Open and reproducible neuroimaging: from study inception to publication. *NeuroImage*, 263.

[9] Levitis, E., van Praag, C. G., Gau, R., Heunis, S., DuPre, E., Kiar, G., ... Finc, K., ... & Maumet, C. (2021). Centering inclusivity in the design of online conferences. *GigaScience* 10(8).

[10] Gut, M., Binder, M., Finc, K., Szeszkowski, W. (2021). Brain activity underlying response induced by SNARC-congruent and SNARC-incongruent stimuli. *Acta Neurobiologiae Experimentalis* (accepted).

[11] Finc, K., Bonna, K., He, X., Lydon-Staley, D. M., Kühn, S., Duch, W., & Bassett, D. S. (2020). Dynamic reconfiguration of functional brain networks during working memory training. *Nature Communications* 11, 2435.

[12] Esteban, O., Ciric, R., Finc, K., Blair, R. W., Markiewicz, C. J., Moodie, C. A., ... & Gorgolewski, K. J. (2020). Analysis of task-based functional MRI data preprocessed with fMRIPrep. *Nature Protocols*, 15(7), 2186–2202.

[13] Dreszer, J., Grochowski, M., Lewandowska, M., Nikadon, J., Gorgol, J., Bałaj, B., Finc, K., ... & Piotrowski, T. (2020). Spatiotemporal complexity patterns of resting-state bioelectrical activity explain fluid intelligence: Sex matters. *Human Brain Mapping.*

[14] Bielczyk, N. Z., Ando, A., Badhwar, A., ... Finc, K., ... OHBM Student and Postdoc Special Interest Group. (2020). Effective self-management for early career researchers in the natural and life sciences. *Neuron* 106(2), 212–217.

[15] Asanowicz, D., Gociewicz, K., Koculak, M., Finc, K., Bonna, K., Cleeremans, A., & Binder, M. (2020). The response relevance of visual stimuli modulates the P3 component and the underlying sensorimotor network. *Scientific Reports*, 10(1), 1–20.

[16] Thompson, W. H., Kastrati, G., Finc, K., Wright, J., Shine, J. M., & Poldrack, R. A. (2020). Time-varying nodal measures with temporal community structure: A cautionary note to avoid misinterpretation. *Human Brain Mapping*, 41(9), 2347–2356.

[17] Bonna, K.*, Finc, K.*, Zimmermann, M., Bola, Ł., Mostowski, P., Szul, M., Rutkowski, P., Duch, W., Marchewka, A., Jednoróg, K., Szwed, M. (2019). Early deafness leads to re-shaping of global functional connectivity beyond the auditory cortex. *Brain Imaging and Behaviour* 15(3), 1469–1482. *equal contribution.

[18] Naumczyk, P., Sawicka, A. K., Brzeska, B., Sabisz, A., Jodzio, K., Radkowski, M., Czachowska, K., Winklewski, P. J., Finc, K., et al. (2018). Cognitive Predictors of Cortical Thickness in Healthy Aging. In: *Advances in Experimental Medicine and Biology.* Springer, New York.

[19] Burzynska, A. Z., Finc, K., Taylor, B. K., Knecht, A., Kramer, A. F. (2017). The dancing brain: Struc-

tural and functional signatures of professional dance training. *Frontiers in Human Neuroscience* 11, 566.

[20] Binder, M., Gociewicz, K., Windey, B., Koculak, M., Finc, K., Nikadon, J., Derda, M., Cleeremans, A. (2017). The levels of perceptual processing and the neural correlates of increasing subjective visibility. *Consciousness and Cognition* 55, 106–125.

[21] Finc, K., Bonna, K., Lewandowska, M., Wolak, T., Nikadon, J., Dreszer, J., Duch, W., Kühn, S. (2017). Transition of the functional brain network related to increasing cognitive demands. *Human Brain Mapping* 38(7), 3659–3674.

Chapters in books

[1] Finc, K. (2023). Network Neuroscience Methods for Investigating Brain Plasticity. In: *The Oxford Handbook of Cognitive Enhancement and Brain Plasticity.*

Contribution to Open-Source Software

fMRIDenoise	Automated denoising and quality control of functional connectivity data. github.com/compneuro-ncu/fmridenoise developer.
fMRIPrep	A robust preprocessing pipeline for fMRI data. github.com/poldracklab/fmriprep contributor.
FitLins	Fitting linear models to BIDS datasets. github.com/poldracklab/fitlins contributor.
Nilearn	Machine learning for neuroimaging in Python. github.com/nilearn/nilearn contributor.

Projects & Grants

2026 – present	Researcher, *EUROpest: A Novel Understanding of Pandemic Disease in Preindustrial Europe (1300–1800)*, ERC Synergy Grant. PI: Adam Izdebski.
2024 – 2025	Principal Investigator, *Exploring the Dynamic Interplay of Brain-Body Network: Advancing Research through Modern Equipment Acquisition*, Excellence Initiative Research University. Budget: 360 000 PLN.
2024 – 2025	Principal Investigator, *Innovative Polyrhythmic Training in Virtual Reality*, Academy of Innovation Leaders. Budget: 30 000 PLN.
2023 – 2024	Co-Principal Investigator, *e-ReproNim: A shared approach to teaching open and reproducible neuroimaging*, Network of European Neuroscience Schools.
2020 – 2021	Principal Investigator, *Modern equipment for audio-visual stimulation and collecting behavioral responses dedicated to fMRI*, Excellence Initiative – Research University, Ministry of Science and Higher Education. Budget: 500 000 PLN.
2019 – 2020	Principal Investigator, *Gamified neurocognitive tests adjusted to fMRI* (KUBUS 2.0), Ministry of Science and Higher Education.
2019 – 2020	Researcher, *Neuronal correlates of reinforcement learning. The role of individual differences in learning strategies* (DSM grant), Faculty of Physics, Astronomy and Informatics NCU. PI: Kamil Bonna.
2018 – 2022	Principal Investigator, *Dynamics and plasticity of the human connectome related to higher cognitive functions*, computational grant, Wroclaw Centre for Networking and Supercomputing.
2018 – 2022	Co-Investigator, *The comprehensive study on the brain basis of low numeracy skills and the behavioral and neuroplastic changes evoked by training of spatial-numerical association* (SONATA BIS, NCN 2017/26/E/HS6/00033). PI: Małgorzata Gut.
2016 – 2018	Principal Investigator, *Temporal dynamics of functional connectivity changes induced by cognitive training. The role of individual differences* (PRELUDIUM 9, NCN 2015/17/N/HS6/03549). Supervisor: Simone Kühn. Budget: 149 642 PLN.
2014 – 2016	Researcher, *Dynamic neural correlates of consciousness as a function of the level of processing* (HARMONIA 4, NCN 2013/08/M/HS6/00004). PI: Marek Binder.
2014 – 2016	Researcher, *Brain correlates of normal cognitive ageing* (PRELUDIUM 5, NCN 2013/091N/HS6/O2634). PI: Patrycja Naumczyk.

Scholarships & Awards

2022	Nominated for the TOP 100 Women in AI in Poland list, Perspectives Foundation.
2021	Jerzy Konorski Distinction for the Best Scientific Publication in Neurobiology, for *Dynamic reconfiguration of functional brain networks during working memory training* (*Nature Communications*, 2020).
2021	Scholarship of the Minister of Education and Science for outstanding young scientists.
2021	Bekker Scholarship, Polish National Agency for Academic Exchange (NAWA), for postdoctoral internship at Max Planck Institute for Human Development, Berlin.
2020, 2023	Lindau Fellowship (Lindau Nobel Laureate Meeting participation).
2020	ReproNim/INCF 2020–2021 Training Fellowship, Center for Reproducible Neuroimaging Computation and INCF.
2018	START travel grant for 1-month study visit in the Poldrack Lab at Stanford University (16 000 PLN).
2018	START Scholarship for outstanding young scholars, additionally increased by the prof. Barbara Skarga scholarship for interdisciplinary research, Foundation for Polish Science (36 000 PLN).
2017	ETIUDA scholarship for PhD candidates (88 838 PLN), National Science Centre (2017/24/T/HS6/00105).
2017	I Prize (Best Oral Presentation Award), *Dynamics and plasticity of functional networks in the human brain*, Neuromania, Toruń.
2014	Erasmus+ Scholarship, internship at Max Planck Institute for Human Development, Center for Lifespan Psychology, Berlin.
2014	Award for the Best Master Thesis, Faculty of Humanities, Nicolaus Copernicus University.
2014	Award for the Best Graduate, Faculty of Humanities, Nicolaus Copernicus University.
2014	Laureate of the regional “Student Nobel” contest.
2014	Entry in the Golden Book of the Best Students, Faculty of Humanities, Nicolaus Copernicus University.
2013–2014	Scholarship of the Minister of Science and Higher Education for outstanding academic performance.
2013	I Prize (Best Poster Award), *Brain in the mourning: neurobiological correlates of resilience*, VIII International Scientific-Educational Conference, Medical University of Białystok.
2011–2014	Rector Scholarship for the best students, Nicolaus Copernicus University.
2010–2011	Scholarship for high academic performance, Nicolaus Copernicus University.

Conference Presentations

[1] Finc, K. (2021). Dynamics and Plasticity of the Human Brain Network During Cognition. Lindau Nobel Laureate Meeting, Next Generation Science Session. *Talk.*

[2] Finc, K. (2021). Understanding the brain’s ecosystem. Neurohackator, Toruń. *Invited talk.*

[3] Finc, K. (2020). Open and Reproducible Neuroimaging, University of Oldenburg. Introduction to fMRI data analysis in Nilearn. *Invited talk.*

[4] Finc, K., Bonna, K., Chojnowski, M. (2019). fMRIDenoise: tool for automatic denoising and quality control of functional connectivity data. INCF Neuroinformatics, Warsaw. *Demo & poster.*

[5] Finc, K., Bonna, K. (2019). fMRIDenoise: automated denoising strategies comparison and quality control of functional connectivity data. Annual Meeting of the Organization for Human Brain Mapping, Open Science Room, Rome. *Talk.*

[6] Finc, K., Bonna, K., He, X., Lydon-Staley, D. M., Kühn, S., Duch, W., & Bassett, D. S. (2019). Dynamic functional network reconfiguration during 6-week working memory task training. OHBM Annual Meeting, Rome. *Poster.*

[7] Finc, K., Bonna, K., Kosik, K., Duch, W., Kühn, S. (2018). Dynamics and plasticity of functional brain networks. University of Washington eScience Institute, Seattle. *Poster.*

[8] Finc, K., Bonna, K., Kosik, M., Duch, W., Kühn, S. (2018). Ongoing dynamics of functional brain network changes during 6-week working memory training. OHBM Annual Meeting, Singapore. *Poster.*

[9] Finc, K., Kosik, M., Bonna, K., Duch, W., Kühn, S. (2018). Task-based Functional Network Changes Following 6-week Working Memory Training. NEURONUS 2018 IBRO&IRUN Neuroscience Forum, Kraków. *Poster.*

[10] Bonna, K., Finc, K., Boła, Ł., Zimmermann, M., Mostowski, P., et al. (2018). Various aspects of compensatory plasticity during resting-state in early deaf adults. NEURONUS 2018, Kraków. *Poster.*

[11] Finc, K., Bonna, K., et al. (2017). Temporal dynamics of functional network changes following 6-week working memory training: preliminary results. Aspects of Neuroscience, Warsaw. *Poster.*

[12] Finc, K., Bonna, K., Lewandowska, M., Wolak, T., Nikadon, J., Dreszer, J., Duch, W., Kühn, S. (2017). Default Mode Network Role in Global Workspace Formation During Increasing Cognitive Demands. Keystone Symposia: Connectomics, Santa Fe. *Talk & poster.*

[13] Bonna, K., Finc, K., Duch, W. (2017). Discovering generative model of human connectome by symbolic regression. Keystone Symposia: Connectomics, Santa Fe. *Poster.*

[14] Bonna, K., Finc, K., et al. (2016). The relationship between whole-brain modularity of the functional network and behavioral performance during a working memory task. OHBM Annual Meeting, Geneva. *Poster.*

[15] Finc, K., Bonna, K. (2016). Functional network reconfiguration related to increasing cognitive effort. NEURONUS 2016 IBRO&IRUN Neuroscience Forum, Kraków. *Talk.*

[16] Finc, K., Bonna, K., et al. (2015). Frontoparietal and default mode network functional connectivity changes during increased load of working memory task. Aspects of Neuroscience, Warsaw. *Poster.*

[17] Bonna, K., Finc, K., et al. (2015). Global modularity changes during increased load of working memory task. Aspects of Neuroscience, Warsaw. *Poster.*

[18] Finc, K., Nikadon, J., et al. (2015). Resting state functional connectivity predicts BOLD activity during working memory task. NEURONUS 2015, Kraków. *Poster.*

[19] Finc, K., Szymaszek, A., Dreszer-Drogorób, J. (2014). Does improvement in temporal information processing underlie cognitive benefits after action video game training? FENS Forum of Neuroscience, Milan. *Poster.*

[20] Finc, K. (2013). The stability-plasticity dilemma in sleep and memory context. Philosophy of Neuroscience, Rostock. *Talk.*

[21] Finc, K., Goraczewski, Ł., Wójcik, A. (2013). Games for brains: a new way of maximizing motivation in education. California Cognitive Science Conference, Berkeley. *Poster.*

[22] Finc, K. (2013). Brain in the mourning: neurobiological correlates of resilience. VIII International Scientific-Educational Conference, Białystok. *Poster.*

Teaching Experience

2023, 2025	Designing Responsible Research for Cognitive Science, laboratory (30h), Nicolaus Copernicus University.
2022 – present	Network Neuroscience, laboratory (30h), Nicolaus Copernicus University.
2021 – 2022	Running a reproducible research project, laboratory (30h), Nicolaus Copernicus University.
2021 – 2022	Neuroscience of perception and attention, lectures (30h), Nicolaus Copernicus University.
2019 – 2022	Fundamentals of fMRI data analysis, laboratory (30h), Cognitive Science (all years), Nicolaus Copernicus University.
2019/2020	Neurobiology, lectures (30h), Cognitive Science (2nd year), Nicolaus Copernicus University.
2015/2016	Neuropsychology, laboratory (30h), Cognitive Science (1st year), Nicolaus Copernicus University.
2014/2015	EEG workshops, laboratory (30h), Cognitive Science (2nd year), Nicolaus Copernicus University.

Additional Training

2020	Neuromatch Academy (interactive track).

2020	Eastern European Machine Learning Summer School: Deep Learning and Reinforcement Learning.
2020	Certified GitHub Campus Advisor.
2020	Crash Course on Python (Google).
2019	The Virtual Brain Workshop (INCF Neuroinformatics, Warsaw).
2018	Neurohackademy Summer School, University of Washington eScience Institute.
2018	Data Scientist with Python, DataCamp (84h career track, 22 courses).
2018	Data Scientist with R, DataCamp (95h career track, 23 courses).
2016	Exploring the Human Connectome, Dutch Connectome Lab, Utrecht Summer School.
2015	Second Brain Connectivity Course, Neurometrika, Grenoble Institute of Neuroscience.
2015	Research team management, scientific communication, and commercialization of research results (Foundation for Polish Science, SKILLS project).
2014	MRI safety training (Max Planck Institute for Human Development).
2014	Data Management for Clinical Research (Vanderbilt University, Coursera).
2014	R Programming (Johns Hopkins University, Coursera).
2014	fMRI data analysis in SPM (SWPS, Warsaw).
2014	Statistical Analysis of fMRI Data (Johns Hopkins University, Coursera).
2013	Simultaneous EEG and fMRI workshop (Brain Products GmbH, ICNT, Toruń).
2013	Simultaneous EEG and TMS workshop (Brain Products GmbH & MAG & More GmbH, Toruń).
2013	Statistics in Medicine (Stanford University, Stanford Online).
2013	Data Analysis (Johns Hopkins University, Coursera).

Other Scientific Activities

2019	Brainhack Toruń organizer.
2019	Brainhack Warsaw organizer.
2016 – present	Supervisor, Neurotechnology Scientific Students Club, Nicolaus Copernicus University.
2016	fMRI workshop for University of Children, Nicolaus Copernicus University in Toruń.
2015	Workshops "Memory Labyrinth", Nencki Institute of Experimental Biology, Polish Academy of Science, Warsaw.
2014, 2013	Co-organizer, NeuroMania Conference, Nicolaus Copernicus University in Toruń.
2010/2011	President, Students' Cognitive Science Circle, Nicolaus Copernicus University.

Membership

2026 – present	Network Science Society (NetSci).
2019 – present	Neuroinformatics Coordinating Facility (INCF).
2016, 2018, 2019	Organization for Human Brain Mapping (OHBM).

Reviewing Activities

Ad-hoc reviewer Nature Methods, NeuroImage, Cerebral Cortex, Current Biology, Frontiers in Neuroscience, Brain Imaging and Behavior, Psychophysiology, Neuroscience, Musicae Scientiae, Neural Plasticity.

Technical Skills

Programming & Tools	Python, R, MATLAB, SQL, Bash, Git/GitHub, LaTeX, Linux; software development
Data Analysis	Graph theory, Representational Similarity Analysis (RSA), Machine Learning, GLM
Neuroimaging	fMRI, EEG, and simultaneous fMRI–EEG data acquisition and analysis
Communication	Scientific writing, academic presentations, data visualization